A counterexample to the tensor-tubal-norm conjecture

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Abstract

The sum of the Euclidean norms of the singular tubes in the standard t-SVD is not a norm. Two real tensors of size $2 \times 2 \times 2$ have value $\sqrt{5/2}$ each, while their sum has value $2 + \sqrt{2}$. This also disproves the proposed convex-relaxation assertion.

Task 1: the original conjecture

Conjecture 2.4.1 in Zhang, Ely, Aeron, Hao and Kilmer [1], p. 4, concerns

$$F(X) = \mathrm{TTN}(X) = \sum_i \|S(i,i,:)\|_2, \qquad X = U * S * V^{\mathsf{T}}.$$

Here the t-SVD is calculated by Algorithm 1 of that source: take the discrete Fourier transform in the third coordinate, compute the usual matrix SVD of each Fourier slice, and apply the inverse transform. We use the standard nonincreasing order of the singular values in each matrix SVD. The conjecture states that F is a norm and is the tightest convex relaxation of tensor-tubal-rank.

This functional differs from the tensor-nuclear-norm in Theorem 2.4.1 of the same source. The latter sums all singular values of all Fourier slices. The counterexample below concerns the singular-tube functional in the conjecture.

Exact counterexample

For a tensor with two frontal slices X_1, X_2, the Fourier convention in the source gives

$$\widehat{X}_1 = X_1 + X_2, \qquad \widehat{X}_2 = X_1 - X_2.$$

Write $s_{ij} = \sigma_i(\widehat{X}_j)$, where singular values are sorted. The ith diagonal tube of S is

$$\left(\frac{s_{i1} + s_{i2}}{2}, \frac{s_{i1} - s_{i2}}{2} \right). \quad \text{Consequently} \quad F(X) = \frac{1}{\sqrt{2}} \sum_i \sqrt{s_{i1}^2 + s_{i2}^2}. \tag{1}$$

This identity follows directly from the inverse Fourier transform and the Euclidean norm; it involves no numerical approximation.

Theorem 1. *The real tensors $A, B \in \mathbb{R}^{2\times 2\times 2}$ with frontal slices*

$$A_1 = \mathrm{diag}(3/2, 0), \quad A_2 = \mathrm{diag}(1/2, 0),$$
$$B_1 = \mathrm{diag}(1/2, 1), \quad B_2 = \mathrm{diag}(-1/2, 1)$$

satisfy

$$F(A) = F(B) = \sqrt{5/2}, \qquad F(A+B) = 2 + \sqrt{2} > \sqrt{10} = F(A) + F(B).$$

Thus F is neither a norm nor a convex function.

Proof. Their Fourier slices and those of their sum $C = A + B$ are

	$\widehat{X}_1$	$\widehat{X}_2$
A	$\mathrm{diag}(2, 0)$	$\mathrm{diag}(1, 0)$
B	$\mathrm{diag}(0, 2)$	$\mathrm{diag}(1, 0)$
C	$\mathrm{diag}(2, 2)$	$\mathrm{diag}(2, 0)$.

Both A and B therefore have singular-value lists $(2, 0)$ and $(1, 0)$ in the first and second frequency slices. Their diagonal singular tubes are $(3/2, 1/2)$ and $(0, 0)$, proving their asserted values. For C the lists are $(2, 2)$ and $(2, 0)$, so its diagonal singular tubes are $(2, 0)$ and $(1, 1)$. This gives $F(C) = 2 + \sqrt{2}$.

All these lists arise from valid real t-SVDs. For A take identity left and right singular-vector matrices at both frequencies. For B take the coordinate-swap permutation matrix for both left and right vectors at the first frequency, and identity matrices at the second. For C take identity matrices at both frequencies. Inverse Fourier transforms of these real frequency factors produce real tensor factors; frequencywise orthogonality gives t-orthogonality and the frequencywise products reconstruct the displayed tensors. The repeated singular value of $\widehat{C}_1 = 2I$ causes no ambiguity in its singular-value list.

Since both sides are positive, the strict comparison follows by squaring: $(2 + \sqrt{2})^2 = 6 + 4\sqrt{2} > 10$. Finally, F is positively homogeneous by (1). Hence

$$F\left(\frac{A+B}{2}\right) = 1 + \frac{1}{\sqrt{2}} > \sqrt{5/2} = \frac{F(A) + F(B)}{2},$$

which violates convexity as well as the triangle inequality. □

A convex envelope must be convex. Thus this functional cannot be the proposed tightest convex relaxation of tubal rank. Although the source does not specify a bounding set for that assertion, the obstruction persists on every norm ball about zero of positive radius: multiply A, B and their midpoint by the same sufficiently small positive scalar. This observation addresses the usual bounded-domain interpretation; the disproof does not depend on the omission of a bounding set.

The source's distinct tensor-nuclear-norm takes values $3, 3, 6$ on A, B, C, respectively, and is not contradicted. The later conference paper [2], Section 2.4, omits the TTN conjecture and discusses the different tensor-nuclear-norm. Its omission alone is not a proof claim resolving the original conjecture.

Proposed follow-up Tasks 2–5

These are proposed research tasks, not further solved problems.

2. Determine the convex envelope of F on the Frobenius unit ball in $\mathbb{R}^{2\times2\times2}$.

3. Determine the smallest constant K such that $F(X + Y) \le K(F(X) + F(Y))$ for all real $2 \times 2 \times 2$ tensors.

4. Characterize real linear subspaces of $2 \times 2 \times 2$ tensors on which the restriction of F is a norm.

5. Find the convex envelope of tensor-tubal-rank on the Frobenius unit ball for real $2 \times 2 \times 2$ tensors and compare it with the normalized tensor-nuclear-norm.

References

[1] Z. Zhang, G. Ely, S. Aeron, N. Hao and M. Kilmer, *Novel Factorization Strategies for Higher Order Tensors: Implications for Compression and Recovery of Multi-linear Data*, arXiv:1307.0805v3, 2013.

[2] Z. Zhang, G. Ely, S. Aeron, N. Hao and M. Kilmer, *Novel methods for multilinear data completion and de-noising based on tensor-SVD*, CVPR 2014, pp. 3842–3849, doi:10.1109/CVPR.2014.485. Official CVF open-access conference copy inspected.